Worksheet A: Cognitive Biases

Name: _____ Date: _____

Task 1: Warm-up: a fictitious short report

in class, individual work, 5 min.

The following report is fictitious and was written for this exercise.

"Baroque music boosts concentration"

(1) A lecturer who has offered concentration training with music for years wants to show that baroque music improves concentration. (2) She recruits 24 students from her own seminar. (3) All of them first solve tasks without music in the morning and then tasks with music in the afternoon of the same day. (4) The lecturer marks the solutions herself; each sheet states whether it was produced with or without music. (5) Twelve measures of concentration are recorded. One of them shows a significant advantage for the music condition; the other eleven are not reported. (6) Because the result after 24 people was "just short of significance", ten more people were tested. (7) The discussion states that the hypothesis was clear from the outset and confirms the author's many years of practical experience. (8) An earlier study by the author that found no effect was never submitted.

For each number, note whether a bias (a systematic distortion in a particular direction) could be at work there, which one, and in which phase of the research process: planning, data collection, analysis, interpretation or publication.

Task 2: Bias hunt in an article

in class, pairs, 25 min.; finish as homework

Article (chosen by the lecturer): _____

Find at least two places where a bias could have influenced the result. Check, one after the other: Who conducted the study, and with what interest? How were participants recruited and assigned? Who measured, and did that person know the condition? How many measures and analysis variants were there, and are all of them reported? Was there a preregistration? Does the interpretation go beyond the data?

Page / quote	Phase	Bias	Reason: in which direction does it shift the result?	Remedy (used?)

What did the authors do well methodologically?